

Using Behavioral Coding to Identify Cognitive Problems with Survey Questions

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Survey questions, like all scientific measures, must meet certain standards. A reasonable set of standards for survey questions is the following:

- Respondents understand the questions as intended by the researcher.
- Respondents have access to the information needed to answer the questions.
- Respondents can recall or retrieve the information needed to provide the answers.
- Respondents can transform the information they have into the form required to answer the questions.
- Respondents are willing to give accurate answers.

Survey research measurement rests on a model that assumes that a consistently understood set of questions is posed that respondents are able and willing to answer. A crucial responsibility of the survey research methodologist is to design questions that pose tasks that respondents are able to perform and that encourage them to report accurately.

Although most researchers would endorse these standards, survey questions are not usually evaluated in ways that would permit researchers to know if their questions meet these standards. In the past decade, stimulated in part by a conference that brought together cognitive psychologists and survey research methodologists (Jabine, Straf, Tanur, and Tourangeau, 1984), survey researchers have begun to take more concrete steps to evaluate their questions prior to inception of a survey. The three main procedures that researchers undertake to help design and evaluate questions are focus groups, intensive cognitive interviews, and field pretests.

The increased use of focus groups and cognitive tests constitutes a significant methodological development for survey research. However, although these procedures are valuable they do not test questions under conditions that replicate usual survey conditions. Moreover, cognitive testing is still not commonly used. Blair and Presser (1993) report that only a third of all academic survey organizations have ever done any cognitive testing of survey questions. In contrast, some sort of field pretest is widely accepted as a routine part of the survey process. Hence, the field pretest constitutes a critical opportunity to identify problems with survey questions.

The traditional field pretest—in which a reasonably representative set of respondents is interviewed with a near-final survey instrument under realistic conditions—has long been standard in survey research. However, the traditional pretest has several serious limitations. First, interviewer observations are the primary sources of information about question problems. There are reasons to be skeptical of the validity and reliability of such observations. Although interviewers can report on aspects of questions that pose problems for them, they are not necessarily competent by training or experience to identify problems with questions that affect the data from an analytical perspective. When conducting a pretest interview, the interviewer's primary focus is on the interaction with the respondent and this may interfere with observation of what goes on. Moreover, the usual pretest includes only a small number of interviews

conducted by each interviewer, providing an inadequate basis for reliable assessment of question problems. Finally, interviewers are frequently inconsistent in the standards they apply to evaluating questions, resulting in lack of agreement about both the questions that cause the problems and the nature of the problems (Bishoping, 1989; Presser, 1989). Thus, although the strength of a field pretest is the generally realistic conditions under which survey questions are evaluated, its weakness is the extent to which systematic, reliable, and valid findings about the questions themselves emerge.

It is in this context that researchers have begun to explore ways to improve the quality of information gained from field pretests. One promising approach is the use of systematic coding of behavior to identify questions that pose problems. Systematic coding of interviewer and respondent behavior provides an objective, replicable technique for measuring what happens in a survey interview. Problems in comprehension and ability to provide answers show up in the behaviors of both respondents and interviewers in consistent and interpretable ways. This chapter describes uses and methods of behavior coding to evaluate survey questions and, in particular, to identify the cognitive problems they pose.

Background of Behavior Coding

For social scientists interested in communication patterns within dyads or groups and for studies of interaction patterns, observation of behaviors has long been a methodological choice. One of the earliest studies of group behavior was conducted by Lewis Terman (1904). The goal was to identify pupils who were leaders and code their interactions with other students.

Clinical psychologists also developed coding techniques to study the processes and outcome of therapy. Research had been hampered because the only data available were counselors' notes taken during the interview. But as Porter (1942) comments, "No two persons could ever agree as to what happened in an interview, or as to what

they did in interviewing, let alone *what ought to be done*" (see Curran, 1945, p. 7). In the 1940s, Carl Rogers and his students developed techniques for coding counseling interviews based on phonographic recording. The goal was to provide objective information to describe counseling interactions and outcomes in order to understand psychotherapeutic processes better.

Porter (1942) developed a system for coding counselor behaviors that he considered a first effort in examining counseling processes. How did counselors' techniques vary over interviews, between interviewers with the same client, and between clients with different problems? Although Porter's analysis was based exclusively on counselor behavior, other studies soon followed that included coding of client behavior. Studies by Royer (1942), Snyder (1947), and Curran (1945) all examined interaction patterns to learn the effects of various counseling techniques.

As interest in studying groups developed in the 1940s and 1950s, coding behaviors became more frequent and more systematic. For example, educators investigated pupil-teacher interaction patterns to identify effective teaching techniques (Flanders, 1960).

The most widely known and completely developed system for measuring group interaction is Bales's *Interaction Process Analysis* (1950). Bales's objective was to design a general set of coding categories that would be useful to study various components of the structure and processes of small groups. His coding scheme consists of a set of categories based on standardized definitions and coding procedures that permit comparison of interactive patterns across small groups. Categories include codes for positive and negative behaviors between group members as well as those related to the task itself. The Bales system has been used in numerous studies of group organization, structure, and process and has provided data for the development of theories on group leadership, cohesiveness, and productivity (Bales, 1950; Cartwright and Zander, 1968).

One of the first uses of behavior coding in survey interviews was in the early 1960s. Record-check studies had documented signifi-

cant reporting error in health surveys. In an attempt to identify sources of reporting error, trained coders observed and systematically coded the behaviors of interviewers and respondents. The conclusion was that "the major causes of good and poor reporting are probably to be found within the interview itself, particularly in the behavior of the participants" (Cannell, Fowler, and Marquis, 1968).

The development of inexpensive portable tape recorders made it possible to record survey interviews and stimulated studies of interviewer-respondent interaction. Some of the early studies were those of Cannell, Fowler, and Marquis (1968), Marquis and Cannell (1969), Marquis (1971), Marquis and Cannell (1971), Morton-Williams (1977, 1979), Sykes and Morton-Williams (1987), Brenner (1982), Dijkstra and Van der Zouwen (1982), Morton-Williams and Sykes (1984), and Dijkstra, Van der Veen, and Van der Zouwen (1985).

These studies revealed that interviewing techniques and interviewer-respondent behaviors frequently deviated significantly from the expectations of the researcher and from accepted norms of standardized interviewing. The first reaction was to blame the interviewer. Deficiencies in performance were attributed to poor training, inadequate supervision, or simple lack of effort. More careful analysis, however, suggested that much of the difficulty rested with the techniques in which interviewers were trained or with inadequacies of the survey questions. Hyman (1975) many years ago put the burden where it belongs, "The fault sometimes is not in our interviewers but in ourselves."

Not surprisingly, the first practical, routine use of behavior coding in survey research was to monitor interviewer performance. Some survey organizations routinely tape-record a sample of each interviewer's work systematically to evaluate such behavior as reading questions and probing. Using systematic behavior coding as part of interviewer supervision is an important advance over relying on the impressions and variable standards of interviewer supervisors. Procedures for monitoring have been described by Cannell, Lawson,

and Hausser (1975), Cannell and Oksenberg (1988), and Fowler and Mangione (1990).

Fowler and Mangione used behavior coding extensively in their study of interviewers. Interviews were tape-recorded and the behavior of interviewers and respondents was coded in detail. Like others, Fowler and Mangione's initial purpose was to evaluate interviewer performance, the extent to which interviewers were reading questions as worded, probing when needed, and probing in a nondirective fashion. However, in the course of their studies, they noticed that certain questions were consistently more subject to interviewer effects than the average. When they examined the behavior associated with individual questions, they found those most likely to be influenced by interviewers usually required probing to obtain an adequate answer (Mangione, Fowler, and Lous, 1992).

This work is significant in demonstrating the place of behavior coding in evaluating questions. First, the findings constitute a bridge between behavior coding and interviewer-related error. The evidence suggests that designing questions so that they can be answered routinely without interviewer probing constitutes one strategy for reducing interviewer-related error. Second, it is evident that questions themselves have significant effects on the respondent-interviewer interaction. Questions subject to high levels of interviewer error in asking them and in probing tend to be associated with high levels of inadequate answers by respondents. Third, the findings demonstrate that the ability of interviewers to perform in a standardized fashion is limited by the questions they are given. Without adequate questions, interviewers cannot follow acceptable procedures. They must choose between using standard procedures that yield inadequate responses or altering the question—by rewording it or using faulty probes—to obtain a response that meets the question's objective. Using behavior coding for monitoring interviewer performance frequently reveals that interviewer errors do not reflect ignorance of techniques or carelessness but rather are attempts to compensate for poor questions.

Thus studies of interviewer performance revealed the reasons behind interviewer deviation from acceptable techniques and laid the groundwork for the use of behavior coding to identify problems with questions during pretests.

Examples of Behavior Coding

The development of a survey instrument progresses through several stages. The final stage is to test the instrument under conditions comparable to those that will be used in the actual survey.

The purpose of the pretest is to evaluate individual questions and procedures for delivering the questions, identifying those that make it difficult for interviewers to administer or for respondents to answer. Pretests typically include twenty-five to seventy-five interviews, which are done by a few interviewers who then meet with the study director in a debriefing session to discuss problems. As noted previously, question evaluation in such pretests traditionally is based on subjective impressions, and interviewers frequently do not agree on questions that cause problems or on the nature of the problems.

Although it has a critical role in surveys, little research has been conducted on pretesting methods. Recently, however, considerable interest has developed in using behavior coding as an objective, quantified technique for evaluating questions at the pretest stage.

We can describe one of these early experimental studies (Oksenberg, Cannell, and Kalton, 1991).¹ A questionnaire including sixty items was constructed by taking questions from health surveys conducted by governmental and university organizations. The questions used were typical of health surveys, referring to medical care and use of services, health status, health behaviors, and health information. All questions had previously been subjected to usual pretest procedures and administered in major health surveys.

The questionnaire was administered to 164 respondents over the telephone by experienced interviewers using standard pretest

techniques. All interviews were tape-recorded. A coding scheme, building on previous work (Cannell and Oksenberg, 1988; Fowler and Mangione, 1990) was used to describe the behavior during the interview.

The objective was to design a coding system that was as short and efficient as possible to identify the primary question problems. If certain key behaviors occurred in at least 15 percent of the pretest interviews when a question was asked, that question was identified as a potential problem. Table 2.1 provides a summary of results for the sixty questions.

As the table shows, in at least 15 percent of the pretest interviews, 30 percent of the sixty questions (that is, eighteen questions) had a significant level of minor interviewer wording change and 5 percent were changed in major ways. Twenty-five percent of the questions had high levels of respondent requests for clarification. The most prevalent problem indicator was inadequate answers. Almost 60 percent of the questions produced an initial inadequate answer 15 percent of the time or more.

Table 2.1. Problem Indicators from Behavior Coding of Questions.

Problem indicator	Percent of questions that evoked behavior at problematical levels ¹
<i>Interviewer question-reading method</i>	
S Slight change	30
M Major changes	5
<i>Respondent behavior</i>	
1. Reading interruption	8
2. Clarification request	25
3. Inadequate answer	60
4. "Don't know"	5

Source: Adapted from Oksenberg, Cannell, and Kalton, "New Strategies for Pretesting Survey Questions," *Journal of Official Statistics*, 7(3), 1991, pp. 349-365.

Note: Results shown are based on sixty interviews.

¹The behavior occurred in 15 percent or more of pretest interviews.

Table 2.2 presents examples of codes for three questions, illustrating different patterns of problem indicators. The first question asked the purpose of a recent visit to a medical provider. The question was simple. Few respondents had problems in answering; none of the problem behavior codes were used in more than 5 percent of the interviews.

In contrast, Question 2, which was on costs, presented many problems. The problems were reflected in high rates of question rewording, respondent interruptions, and inadequate answers.

The last question showed a very high proportion (nearly 90 percent) of inadequate responses. The problem was a lack of under-

Table 2.2. Percent of Problem Indicators for Selected Questions.

Problem Indicator	Percent of problems per question		
	Question 1	Question 2	Question 3
<i>Interviewer question-reading method</i>			
S Slight change	2	30	3
M Major changes	3	17	2
<i>Respondent behavior</i>			
1. Reading interruption	0	23	0
2. Clarification request	2	10	3
3. Inadequate answer	5	17	87
4. "Don't know"	0	8	12
<i>The questions:</i>			
1. What was the purpose of that visit (to a health care person or organization)?			
2. How much did you pay or will you have to pay out-of-pocket for your most recent visit? Do not include what insurance has paid for or will pay for. If you don't know the exact amount, please give me your best estimate.			
3. When was the last time you had a general physical examination or checkup?			

Source: Adapted from Oksenberg, Cannell, and Kalton, "New Strategies for Pretesting Survey Questions," *Journal of Official Statistics*, 7(3), 1991, pp. 349-365.

standing of the concept "general physical exam and checkup." More important, the question did not communicate the kind or form of answer the people were being asked to give. Asking people "when" something happened does not specify whether they should give a date or the number of months or years ago or their own age at the time, nor does it specify how precise the answer should be. Because respondents have different opinions about what they think is wanted, the result is many inadequate answers.

Tables 2.1 and 2.2 reveal that many questions cause serious difficulty for both interviewers and respondents. We mention again that these questions were taken from major surveys and had been pretested with conventional techniques. Yet behavior coding indicated serious problems that the original pretesting presumably did not reveal.

Evidence of the reliability of the method comes from this same study. The same sixty questions were tested by interviewing staffs from two organizations with comparable samples.

On a question by question basis, when the rates at which interviewers misread questions, respondents asked for clarification, and respondents provided inadequate answers to questions were compared across the two staffs, the rates correlated in a .6 to .8 range, demonstrating that questions have predictable, consistent effects on interviewer and respondent behavior that are independent of the survey organization or the particular interviewers involved.

Another analysis that provided some evidence of the validity of diagnoses based on behavior coding was reported by Fowler (1992). Seven questions in the survey instrument were identified as having key terms that were poorly defined based on high rates of requests for clarification and inadequate answers. The questions were redesigned to provide better, more complete definitions of what was required and the survey was repeated. When the responses from the initial and the redesigned questions were compared, two kinds of changes were consistently found. First, there were many fewer requests for clarification and lower rates of inadequate answers. Second, for five of the seven questions, the content of the responses

changed significantly in a predicted direction. For example, when a question about the frequency of exercise was better defined to make it clear that walking for exercise was included, the rate at which respondents reported exercising significantly increased, as would be predicted. This helps build the case that the analysis of behavior coding produces meaningful conclusions and that those conclusions, when reflected in question redesign, can improve the validity of survey data.

The previous analyses have examined individual questions as if each were an independent event. The following analysis examines questions in relation to each other and to factors that influence the behavior of interviewer and respondent (Oksenberg, Cannell, and Blixt, 1992). A sample of 250 interviews, part of a national survey on the uses and costs of medical services, were tape-recorded and behavior coded. Only about 33 percent of all questions were asked verbatim. About 25 percent more were asked with changes judged not to alter the intent of the question. The balance of the questions (about 40 percent) were asked with potentially biasing changes.

The complexity of the task proved to be a major correlate of the interviewer's question-asking behavior. In this interview schedule, large numbers of repetitive questions made for a tedious, demanding interview. Questions were repeated for each person in the family and for each event reported. Interviews were lengthy and placed considerable demands on respondents' memories and patience. Interviewers altered questions or omitted them entirely to make the interview less repetitious, more pleasant, less demanding, and more conversational and to create good relationships with the respondents.

A sequence of questions on disability will illustrate the problem. Table 2.3 shows that in 72 percent of the interviews, the lead question was either asked as written, reworded without apparent bias, or reworded to verify information obtained earlier. In the remaining 28 percent of the interviews, the question was seriously reworded or not asked at all and an answer was inappropriately inferred from previous answers.

Table 2.3. Percent of Interviews in Which Each Question Was Asked Acceptably.

Percent asked acceptably	Question series
72	Including any days spent in a hospital, between [START DATE] and [END DATE] did illness or injury keep [PERSON] from work more than half of the day, not counting work around the house?
THOSE ANSWERING "YES" WERE ASKED THE FOLLOWING QUESTIONS:	
42	What were the dates that [PERSON] missed more than half a day from work between [START DATE] and [END DATE]?
15	How many actual work days was that between [FIRST DATE] and [SECOND DATE]?
47	What condition caused [PERSON] to miss work from [DATE] through [DATE]?
48	Was there any other condition?
30	Were there any other dates that [PERSON] missed work for more than half a day between [START DATE] and [END DATE]?

Note: The respondent was asked this sequence about each member of the family. The time reference was the last interview, usually about three months.

The next two questions showed a sharp decrease in acceptable techniques; there was a significant increase in omitting the question or erroneously inferring a response. The fourth question introduced a new subtopic and acceptable question handling significantly improved. A follow-up question again demonstrated a decrease in acceptable behavior.

The most significant finding of this analysis is that interviewers frequently do not follow procedures as specified. The researchers created questions defining and describing in detail each variable for which information was needed. Interviewers were trained to ask

questions as worded and the researchers analyzed the data with the assumption that all questions had been administered as specified. Yet the behavior coding revealed that the researchers had designed an instrument that interviewers could not administer in a standardized way.

This survey is not unique. The findings are replicated in other large governmental and university surveys designed to collect descriptive and factual information on varied topics, including health behavior, morbidity, use of health services, financial issues including spending and saving behavior, durable goods purchases, and so on. Behavior coding provides a window into the "black box" of the interview, revealing how questions are administered and the nature of the resulting interviewer-respondent interaction. In short, behavior coding documents the way in which a survey was actually carried out as no other procedure can.

An extension of the behavioral coding technique was an analysis of pretests of seven surveys using questionnaires designed by investigators from various organizations covering a variety of topics. The results of the behavior coding were examined to identify features common to questions that were most problematical (Can-nell, Blixt, and Oksenberg, 1992). Although this analysis is only preliminary, the following are examples of the generalizations that appeared to emerge.

1. Questions that are not read as worded are awkwardly phrased or include words that are difficult to pronounce.
2. Questions that are misread and frequently interrupted often offer dangling explanations at their conclusion.
3. Questions that lead to requests for clarification often require response tasks that do not fit respondents' experience or frame of reference.
4. Questions that require clarification often are vague or contain a poorly defined term or concept.

5. Questions that stimulate inadequate answers often ask for a level of detail that is greater than the respondent can provide.

This list illustrates the potential of behavior coding to provide a basis for question wording principles. The above principles may be too broad to provide useful guidance to question designers, but data from recurrent use of behavior codes can begin to provide better specification and examples based on objective observation.

A Protocol for Behavior Coding

An important practical issue is how best to integrate behavior coding into routine survey procedures. In this section we offer a brief outline of our experience in using behavior coding. As noted, coding behaviors is useful for monitoring interviewer technique and identifying problems with questions in pretests and for making methodological studies of data-collection techniques. For these applications, behavior coding schemes vary according to the purpose. For example, when monitoring interviewer performance, only interviewer behavior is coded, whereas when trying to identify problems with questions, respondent behaviors are more useful. However, interviewer and respondent behaviors are highly interrelated. When a respondent gives an inadequate answer, the interviewer usually probes.

A few key behaviors are particularly useful in identifying question problems, including comprehension, lack of shared understanding of terms or concepts, communication difficulties, or cognitive difficulty in processing information. Over a series of studies, a coding system has been developed with a minimal number of diagnostic codes for identifying questions that are problematical for interviewers or respondents.

Table 2.4 provides an example of a coding scheme of interviewer and respondent behaviors.

Behavior coding can be readily incorporated into any telephone

Table 2.4. Examples of Behavior Codes.

Interviewer question-reading codes	
E Exact	Interviewer reads the question exactly as worded.
I Slight change ¹	Interviewer reads the question with minor changes that do not alter the question meaning.
S Major change ¹	Interviewer changes the question so that the meaning is altered; Interviewer does not complete question reading.
Respondent behavior codes	
1. Interruption ¹	Respondents interrupt initial question reading with an answer.
2. Clarification ¹	Respondent asks for repeat or clarification of question or makes statement indicating uncertainty about question meaning.
3. Adequate answer ¹	Respondent gives answer that meets question objective.
4. Qualified answer ^{1,2}	Respondent gives answer that meets question objective but is qualified to indicate uncertainty about accuracy.
5. Inadequate answer ¹	Respondent gives answer that does not meet question objective.
6. "Don't know" ¹	Respondent gives a "don't know" or equivalent answer.
7. Refusal to answer ¹	Respondent refuses to answer the question.

¹Indicates potential problem with the question.

²In some coding schemes, these two codes are omitted.

or personal interview pretest. The effect of questions on behavior is reliable enough that thirty pretest interviews usually is sufficient to produce meaningful results.

Behavior coding can be done by an observer during face-to-face interviews or by a monitor listening on an extension phone during telephone interviews. A more efficient procedure, however, is to code from tape recordings. Tapes are available when wanted so the coder does not have to be available at the interviewer's convenience. A more significant advantage is that the coder can stop the tape to review content or make brief notes of problems.

Coders need to have basic interviewer training in order to understand the principles and "house rules" of question asking including how and when to probe and acceptable techniques for responding to requests for clarifications and "don't know" responses. With trained coders, the intercoder reliability is quite satisfactory. Measured by the kappa statistic, agreement ranges from .75 to .90.

Training for this task is similar to any other coder training. Short excerpts of recorded interviews are played and discussed until general consensus is obtained. Following this, segments of interviews are independently coded and results compared.

Codes can be recorded on coding sheets. Coding sheets simply include the code categories along the top and question numbers down the left margin. Coding is done by a check mark in the appropriate box. The results can be summarized by counting the numbers of checks in each category for each question.

A simpler and faster method is to enter the codes directly into a computer database, which allows the results to be tabulated at any point during production coding and permits analysis to begin immediately following completion of coding.

A key issue in the use of behavior codes is how best to diagnose the basis for the problem. Many times the cause is readily apparent upon examination of the question in relation to the problem indicated. In other cases, inference from a problem indicator to the cause is more difficult.

The simplest protocol is to involve behavior coders and interviewers directly in the diagnostic process. A question by question summary of the frequencies of each coded behavior is tabulated before a debriefing meeting. The behavior coding results themselves become a subject for discussion, with input from coders and interviewers. Generally, coders (more than interviewers) can recall the origins of most of the troublesome behaviors. Work is now under way to develop and test codes for the nature of the problem, to facilitate coder identification of the reasons for problems, and to help make problem diagnosis more systematic.

Table 2.5 summarizes the primary sources of question problems that emerge from work accomplished to date. It will be noted, however, that the same problem can affect behaviors in more than one way, and there are multiple possible causes for the same behavior.

Finally, it should be noted that the function of the behavior coding and the input of interviewers and coders is to diagnose problems, to help the investigators understand what it is about the question that interferes with a positive interaction and a productive question and answer process. It is the task of the investigator to decide whether and how to revise the question to eliminate the problem.

How Behavior Coding Relates to Other Question Evaluation Techniques

As noted at the beginning of this chapter, focus group discussions, cognitive interviews, and behavior coding of field pretest interviews are all strategies that can contribute to the question design and evaluation process. Given the ubiquitous constraints of time and money that researchers face with every survey, the issue becomes the extent to which focus groups, cognitive interviews, and field pretests are redundant. What would a parsimonious and optimal protocol for question evaluation look like?

Focus groups can provide useful information about vocabulary,

Table 2.5. Common Question Problems Affecting Respondent and Interviewer Behavior.

Behavior	Common problems
Interviewer misreads questions	• Awkward wording (difficult to pronounce). • Introductions that interviewers find superfluous or do not fit context. • Dangling clauses at end of question. • Missing explanation of response task. • Unclear, undefined terms. • Unclear response task. • Poor question order; respondents cannot recall response alternatives.
Respondent requests clarification	• Wording results in a complete question before the question is finished. • Respondent does not realize response alternatives will be given.
Respondent interrupts	• Meaning of question is unclear; undefined terms. • Unclear response task or poor question order so respondent cannot recall response alternatives. • Response task does not fit answers. • Response task is difficult, requires effort to recall or asks too much detail.

concepts, and the realities about which people are reporting. However, they are not usually designed to provide specific tests of question wording or of the cognitive burden placed on respondents. Cognitive interviews provide valuable techniques for evaluating respondent comprehension of individual questions, revealing how they perform the tasks that are specified in the questions (Lessler, 1987; Lessler and Tourangeau, 1989; Forsyth and Lessler, 1991a). The limits of cognitive interviews tend to be the small, unrepresentative samples and the unique conditions under which they occur.

Royston (1989) reports some anecdotal analyses that suggest that some problems that do not show up in the cognitive laboratory can become problems in a normal field setting. In contrast, only occasionally do questions that were problems in cognitive laboratories fail to pose difficulties under more realistic survey conditions.

Two studies to date compare the results of cognitive interviews, focus groups, and behavior coding of field pretests. In general, the results of cognitive tests and focus groups were complementary to results from behavior coding of field pretests rather than redundant. The problems identified overlapped but were far from identical (Presser and Blair, 1993; Fowler and Roman, 1992). The quantitative nature of the behavior coding results was clearly a distinctive advantage of that procedure compared with the others.

Conclusion

Behavior coding has been demonstrated to be a valuable technique for studying problems of survey questions under realistic data-collection conditions. One of its primary values is the production of data that are comparable across different surveys. Coding the behavior in presurvey evaluation interviews and during production interviews can reveal systematic relationships between question characteristics and resulting problems. By providing a window into the actual process of data collection, behavior coding offers unique potential for improving the quality of generalizations about question design and for measuring the contribution of the data-collection process to overall survey error.

The potential of behavior coding to contribute to the study of survey methods needs to be further developed. On the one hand, we need studies of the question characteristics that lead to problems reflected in behavior coding. On the other hand, we need studies of the relationship between problems indicated by behavior coding and independent measures of error in survey data (Mangione, Fowler, and Louis, 1992).

An important component missing from most methodological reports is an evaluation of questions and the quality of the information they produce. Although there are well-established standards for reporting sampling error and response rates, there are no comparable quantitative measures of response error. We think behavior coding has the potential to help fill the gap. Studies demonstrate that questions frequently fail to meet quality standards and fail to be administered according to quality standards.

Users of survey data lack information about the quality of the data-collection process in general and the quality of the questions in particular. Behavior coding with its quantitative nature and its demonstrated relationship to key measures of data quality can provide indicators to readers on both subjects.

From a total survey design perspective (Fowler, 1993; Groves, 1989), it can be argued that improving the design and evaluation of survey questions is the most important methodological issue that survey research practice faces. A protocol consisting of focus groups, cognitive interviews, and a field pretest that includes behavior coding should be part of the development of every survey instrument. Unfortunately, evidence from our work suggests that many survey questions are seriously underevaluated. Certainly, behavior coding has great potential in producing systematic, replicable, and effective presurvey evaluation of questions.

The timing of evaluation matters too. The most common use of behavior coding today is to evaluate questions at the final pretest stage, just before a survey is to begin. However, this often is too late in the process. Many question problems are simply the result of poor wording that requires editing but some involve ambiguity in the concepts to be measured. Furthermore, problems may involve not just one but an entire sequence of questions. If problems with the basic objectives and concepts covered in a survey are discovered at the final pretest stage it may be too late for extensive question redesign. For this reason, to benefit fully from the results, a series of question evaluation steps that begins well before the survey will be launched is needed.

The coding scheme described in this chapter is designed to identify problems with questions in an efficient and cost-effective manner. Diagnosing the reasons for the problems requires an inferential step and is one of the areas most in need of further work. We are now attempting to develop a scheme for coders to record their assessments of the basis for the question problems they observe. Informal testing to date indicates that coders can agree in their assessments of the reasons for the problems observed through behavior coding.

Current codes are focused largely on cognitive problems: question comprehension, concept clarity, and response task difficulty. This work needs to be extended to identify questions that have adverse effects on respondents' feelings and motivations. The questionnaire must not only communicate the question objectives but also assist the interviewer in maintaining attention, interest, and motivation. So far, there are no systematic strategies for evaluating the way survey questions affect these aspects of the interview.

Finally, we would be remiss if we neglected to mention behavior coding's potential to identify problems that affect interviewer-respondent interaction. A survey interview is not simply a set of statistical tasks or cognitive tasks, it also is an interpersonal event (Schaeffer, 1991; Schaeffer and Maynard, Chapter Four, this volume; Suchman and Jordan, 1990). The interview schedule is a protocol for the question and answer process. When the interview schedule is poorly designed it can have a negative effect on the respondents' orientation to the interview, which in turn can affect the quality of data provided. Behavior coding of tape-recorded interviews has unique potential for identifying questions or segments of an interview schedule that hinder the respondent and interviewer from working together in a smooth, standardized way. Determining the adequacy of a survey instrument from this perspective is another critical goal of presurvey evaluation.

Thus, at this time, techniques for using behavior coding to identify cognitive problems posed by questions are ready for routine use. There still is work to be done to make it easier to interpret and use

the results for question redesign. However, we believe that in the long run behavior coding may contribute to the evaluation and improvement of virtually all parts of the survey data-collection process.

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Questionnaire Pretesting

Computer-Assisted Coding of Concurrent Protocols

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Over the past six years, GTE has worked to enhance the quality of the information it obtains through its surveys, focusing on making improvements to its customer satisfaction questionnaires. Our experiences have led to the development of a new pretesting methodology that identifies respondents' cognitive difficulties as they form answers to survey questions. This methodology entails the computer-assisted coding of concurrent verbal protocols elicited during pretest interviews. Our prior research has shown that this methodology can be used to identify and improve defective questions.

This chapter elaborates on two aspects of our pretest methodology. It describes techniques for eliciting concurrent verbal protocols in pretest interviews and an extension to our automatic coding scheme designed to measure respondents' cognitive difficulties answering questions about low involvement, low frequency events. The study context is a pretest of a consumer survey concerning local telephone service. The study focuses on questions about low involvement telephone experiences—such as calls to directory assistance—that are likely to be relatively low frequency events for most customers.

The chapter is organized in the following way. The first section briefly describes existing pretesting methods and summarizes our development of a pretesting method based on computer-assisted coding of concurrent protocols. The two subsequent sections